

Learning Geometric Knowledge from Text for Effective Geometry-Free Adsorption Configuration Screening

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Abstract—Identifying the optimal adsorption configuration for a given adsorbate–catalyst system is essential for catalyst design. Traditional methods rely on density functional theory (DFT), which is computationally expensive. Geometric deep learning has emerged as a more efficient alternative, yet its high computational cost in processing complex 3D structures remains a bottleneck for high-throughput screening. By contrast, language models operate directly on textual descriptors, offering faster inference but at the cost of discarding explicit 3D geometric information and sacrificing prediction accuracy. To achieve both the efficiency of language models and the accuracy of geometric models, we propose an optimal transport framework that learns a mapping from textual descriptors to geometric structures during training, enabling geometry-free inference at test time. Specifically, we formulate the alignment between text and geometry distributions as an optimal transport problem. A refined optimal transport operating at both the group and sample levels establishes a precise one-to-one correspondence from textual descriptors to geometries. Once trained, our model predicts relaxed adsorption energies directly from textual descriptors alone, allowing rapid evaluation across numerous candidate configurations and reliable identification of the optimal one. On the initial-structure-to-relaxed-energy task, our method achieves accuracy on par with or better than state-of-the-art geometric models while running more than $2\times$ faster.

Index Terms—AI for Chemistry, Molecular Property Prediction, Multimodal Learning, and Language Models.

I. INTRODUCTION

Accurately and efficiently identifying the optimal adsorption configuration corresponding to the global minimum in adsorption energy is a cornerstone of rational catalyst design [1]–[7]. For a given adsorbate–catalyst pair, a vast set of candidate structures can arise, distinguished by adsorption sites, molecular orientation, and subtle local geometric arrangements. Conventional identification of the optimal configuration conventionally depends on exhaustive density functional theory (DFT) structural relaxations and energy calculations [8]–[11]. Nevertheless, the formidable computational overhead severely limits its applicability to high-throughput screening workflows. This bottleneck has spurred the rapid advancement of data-driven surrogate models to accelerate computational catalyst exploration [12]–[18].

Machine learning has offered efficient alternatives for adsorption configuration design [19]–[23]. These methods are often

distinguished by how they represent the adsorption system. One family adopts geometric neural networks, which treat the system as an atomistic graph and learn from atomic positions, local neighborhoods, and geometric relationships [17], [24]–[27]. Among others, architectures such as DimeNet/DimeNet++ [28], [29], GemNet-OC [30], [31], and EquiformerV2 [32] further improved performance by explicitly modeling distances, angles, and symmetry-aware interactions. A second line of work leverages large language models by converting the molecular system into a structured text sequence and performing prediction directly from the textual description [13], [33]–[37]. These methods demonstrate the potential to bypass 3D geometries using textual descriptors. For instance, CatBERTa shows that chemically meaningful text descriptors alone can support relaxed adsorption energy prediction [18]. Building on CatBERTa, GAP aligns geometric and textual descriptors via a CLIP-style framework [38], [39]. After alignment, predictions rely solely on textual descriptors without requiring explicit geometric input.

Despite the progress, both paradigms face fundamental trade-offs between accuracy and efficiency. Geometric deep learning models achieve high predictive accuracy by explicitly encoding 3D atomic coordinates and bond geometries, but this requires expensive iterative aggregation processing of atomic graphs and imposes severe computational bottlenecks for large-scale high-throughput screening [5]. By contrast, language models operate directly on chemically structured textual descriptors, bypassing explicit 3D structure processing. This yields faster inference, yet the omission of explicit geometries typically results in lower accuracy and limited generalization to out-of-distribution adsorption systems [18].

Striking a balance between efficiency and accuracy is challenging. Adsorption energy is highly sensitive to subtle local atomic arrangements, yet textual descriptors—such as “top” vs. “bridge” adsorption sites—cannot precisely capture such fine-grained geometric differences. In principle, there exists a one-to-one correspondence between each textual descriptor and its associated geometry. Therefore, to bypass 3D geometry for efficiency while preserving accuracy, a model must learn an instance-level one-to-one correspondence: each textual descriptor of a configuration instance must map precisely and uniquely to its own geometry. This requirement distinguishes our approach from GAP [38], which only aligns geometric and textual descriptors via a CLIP framework—pulling paired text–geometry embeddings closer together without establishing a precise one-to-one mapping from text to geometry.

To address this fundamental trade-off between computational efficiency and prediction fidelity, we propose *Taco*, a multi-modal framework that learns a one-to-one mapping between textual descriptors and their corresponding adsorption geometries. We formulate text-geometry alignment as a distribution matching problem and establish correspondences between semantic and geometric representations through optimal transport. Unlike conventional optimal transport methods that perform alignment between two distributions directly, we propose a Group-Conditioned Optimal Transport (GCOT) framework based on soft cluster assignments in the latent feature space. Specifically, textual and geometric representations are first associated with a set of learnable cluster prototypes through soft assignment, producing group-aware distributions in both modalities. GCOT then performs transport-based alignment at the group level to identify cluster correspondences, followed by instance-level transport within aligned groups to refine sample-wise matching between textual descriptors and their corresponding three-dimensional adsorption structures. This hierarchical alignment mechanism effectively bridges the gap between textual and geometric representations, allowing accurate adsorption energy prediction without requiring explicit geometric inputs during inference. The main contributions of this work are summarized as follows:

- 1) We propose *Taco*, a geometry-free adsorption energy prediction framework that establishes a direct mapping from textual descriptors to geometric latent representations, enabling efficient adsorption property prediction using textual inputs only.
- 2) We develop a novel fine-grained optimal transport for multimodal alignment. By jointly modeling group-level semantic-structural correspondences and instance-level transport relationships, GCOT effectively bridges the representation gap between textual and geometric modalities.
- 3) We formulate the cross-modal alignment problem as a hierarchical optimal transport optimization on the Stiefel manifold and derive a distributed ADMM-based solver that efficiently estimates both cluster correspondences and geometric transformations.
- 4) We provide theoretical analysis of the proposed GCOT framework, including sufficient conditions for cluster correspondence identifiability and perturbation bounds that characterize the quality of the learned cross-modal alignment.
- 5) Extensive experiments on the Open Catalyst benchmark demonstrate that *Taco* achieves competitive or superior prediction accuracy compared with state-of-the-art geometry-based methods, while reducing inference cost by more than a factor of two and eliminating the requirement for explicit geometric structures during deployment.

II. PRELIMINARIES

Problem definition. We have two modalities for representing the adsorption configurations: geometric representations $\mathcal{G}$ and textual descriptors $\mathcal{S}$ [18]. For the textual descriptors, as shown

in Figure 1(a), we include the adsorbate formula Ad , surface elements Su , Miller indices Mi , primary atoms Pr , site types Si , and secondary atoms Se . These attributes are concatenated in a fixed order and tokenized as $\mathcal{S} = \text{Tokenizer}(\text{Ad} \oplus \text{Su} \oplus \text{Mi} \oplus \text{Pr} \oplus \text{Si} \oplus \text{Se})$. Pr and Se provide chemical and site-level cues rather than explicit atom coordinates or atom-level masks.

For the geometric representation, each adsorption system is modeled as an atomistic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, \dots, v_N\}$ denotes the atom set with coordinates $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times 3}$. Under periodic boundary conditions, the edge set is constructed as $\mathcal{E} = \{(i, j, \mathbf{k}) \mid \|\mathbf{x}_i - (\mathbf{x}_j + \mathbf{L}\mathbf{k})\|_2 \leq R_{\text{cut}}\}$, where $\mathbf{L}$ is the lattice matrix, $\mathbf{k}$ is the cell offset vector, and R_{cut} is the cutoff radius. Among the candidate neighbors within R_{cut} , the K nearest neighbors are retained.

For a given adsorbate-catalyst pair, there exists a candidate set of adsorption configurations $\mathcal{C} = \{C_1, C_2, \dots, C_M\}$, which vary in adsorption sites, molecular orientations, and local atomic arrangements. Each configuration C_m can be represented by a geometric graph $\mathcal{G}_m$, a textual descriptor $\mathcal{S}_m$, or both. We aim to train a predictor f_θ to estimate the adsorption energy $\hat{E}_{\text{ads},m}$ for each configuration, using $\mathcal{G}_m$, $\mathcal{S}_m$, or their combination as input. The screening task can be reduced to identifying the configuration with the lowest predicted energy: $C^* = \arg \min_{C_m \in \mathcal{C}} \hat{E}_{\text{ads},m}$. The selected C^* is taken as the most stable adsorption structure among the candidates for the given adsorbate-catalyst system. In the following, we discuss several approaches that can be used for adsorption configuration screening.

Graph-based models. Most adsorption energy prediction models operate on atomistic graphs, using local coordination and geometry as the main input [24], [27], [40]. DimeNet [41] introduces directional message passing that incorporates bond angles into the message construction, while DimeNet++ [42] improves upon it with a faster and more memory-efficient architecture. GemNet [30] extends this direction with two-hop message passing that captures dihedral angles, achieving universal approximation for rotationally equivariant predictions [43]. Equivariant architectures like PaiNN [44] extend invariant message passing to rotationally equivariant representations, enabling the prediction of tensorial properties such as molecular spectra. eSCN [45] incorporates higher-degree tensors into equivariant convolutions efficiently. EquiformerV2 [46] further improves upon Equiformer by replacing $\text{SO}(3)$ convolutions with eSCN convolutions and introducing architectural enhancements. Nevertheless, these models require explicit 3D geometries at test time, which limits their applicability to high-throughput screening.

Text-based models. A different set of approaches avoids full geometric inputs by encoding molecular systems with descriptors or textual representations [47]. Language models pretrained on scientific text, SciBERT [48], BioBERT [49], MatSciBERT [50], MOFormer [51], and Molecular Transformer [52], have shown that textual pretraining captures domain-specific knowledge, with BioBERT demonstrating strong performance in the biomedical domains [49], MatSciBERT achieving competitive band gap prediction with significantly fewer

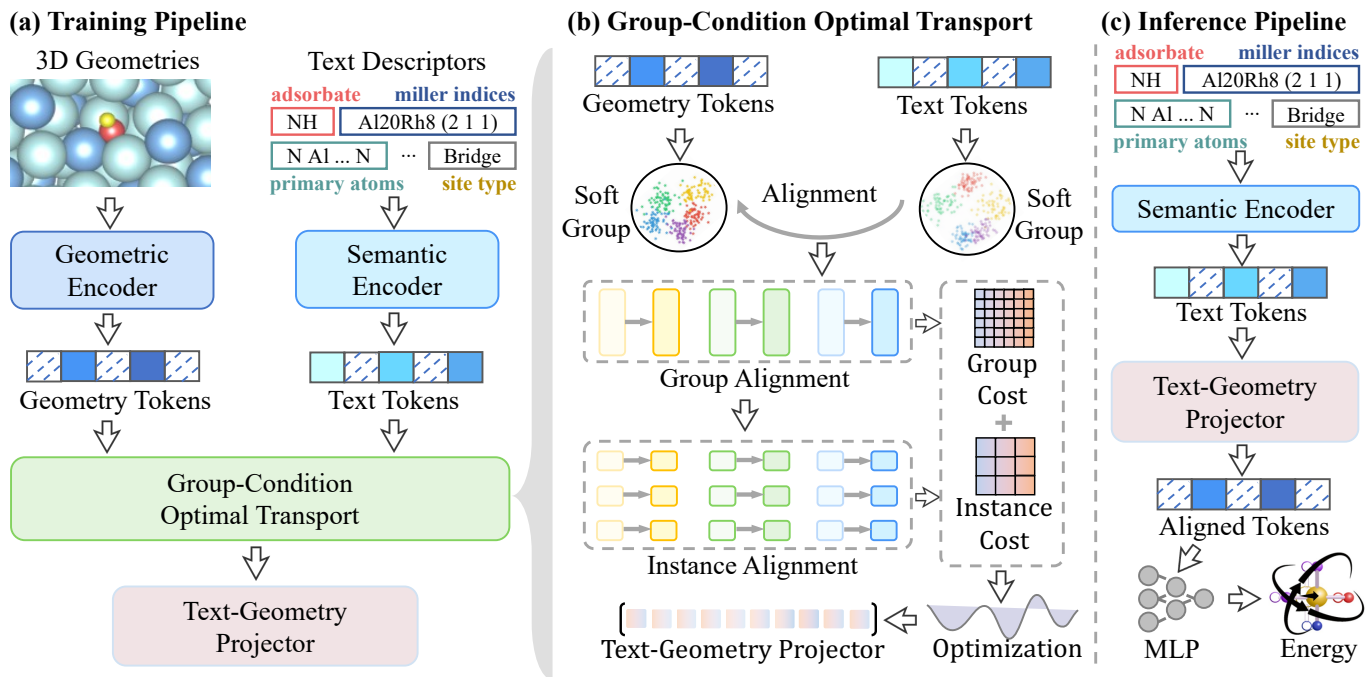


Fig. 1: Overview of the *Taco* framework. (a) Training stage: 3D geometries and text descriptors are encoded separately into tokens, which are aligned via a group-conditional optimal transport module at both the group level (distributions) and sample level (one-to-one correspondence). This learned text-geometry correspondence enables efficient text-only inference. (b) Test stage: A candidate configuration’s geometry is converted to text and fed into the semantic encoder. The learned correspondence then generates geometric tokens for energy prediction, bypassing the high computational cost of geometric neural networks. This allows rapid screening to identify the globally optimal adsorption configuration.

parameters [50], and MOFormer designed for multi-objective peptide optimization rather than property regression [51].

Among these works, one line of studies converts adsorption configurations into structured text sequences, demonstrating that textual descriptors alone can be used for energy prediction. CatBERTa [18] shows that chemically meaningful text descriptors alone can support relaxed adsorption energy prediction. Building on this, GAP [38] aligns geometric and textual descriptors through a CLIP-style contrastive learning framework [39], enabling inference from text alone after alignment.

However, this efficiency comes at the cost of accuracy, as these methods discard the precise 3D geometries. Although GAP pulls paired text–geometry embeddings closer in a shared space, it does not establish a direct, explicit mapping from textual descriptors to their corresponding geometric structures. Consequently, predictive accuracy remains suboptimal. In this work, we aim to learn a precise text-to-geometry mapping that enables geometry-free inference while striking a more favorable accuracy–efficiency balance.

III. METHODOLOGY

A. Overview Pipeline

As shown in Figure 1(a), *Taco* takes two predefined inputs introduced in the problem formulation: the 3D geometric graph $\mathcal{G}$ and textual descriptor $\mathcal{S}$. Concretely, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ encodes atomic coordinates and periodic connectivity with neighbor truncation at K nearest atoms, while $\mathcal{S}$ corresponds to the ordered concatenation and tokenization of seven handcrafted chemical attributes that encode interpretable local environmental information. The model proceeds to fuse these two

modalities, extract physically meaningful local geometric motifs $\mathcal{G}_{\text{local}}$, and ultimately predict adsorption energy following the regression formulation $\hat{E}_{\text{ads}} = f_{\theta}(\mathcal{G}_{\text{local}}, \mathcal{S})$.

At the training stage, the geometries $\mathcal{G}$ and textual descriptors $\mathcal{S}$ are fed into a geometric neural network and a text encoder, producing geometry tokens and text tokens, respectively. These tokens are then aligned via a GCOT module, which operates at two levels: At the group level, it aligns the distribution of textual descriptors with that of geometric structures within each natural group. At the sample level, it establishes an exact one-to-one correspondence between individual text descriptors and their corresponding geometries. By minimizing the optimal transport costs at both group and sample levels, we learn a dedicated projector that maps textual representations into the geometric feature space. This learned projector serves as the core component that enables efficient geometry-free inference using only textual descriptors.

At test time, as illustrated in Figure 1(b), the geometry of a candidate configuration is first converted into a textual description and then fed into the semantic encoder. Subsequently, the pre-learned text-geometry correspondence generates the corresponding geometric tokens, based on which the energy head performs energy prediction. Unlike geometric neural networks, which consume heavy computational overhead for graph structure aggregation, our pipeline enables fast relaxed energy prediction without geometric processing. Consequently, *Taco* allows for rapid screening of adsorption configurations, facilitating efficient and reliable identification of the optimal configuration that achieves the global minimum adsorption energy. Details of the components are provided in the Methods section.

B. Encoders

Geometric encoder. We employ the encoder from EquiformerV2 [32] to map the atomic graph into a symmetry-preserving latent space. To ensure translation invariance, the encoder operates on relative displacement vectors $\mathbf{x}_i - (\mathbf{x}_j + \mathbf{L}\mathbf{k})$, where $\mathbf{x}_i, \mathbf{x}_j \in \mathbb{R}^3$ denote the Cartesian coordinates of atom i and atom j , $\mathbf{L} \in \mathbb{R}^{3 \times 3}$ is the lattice matrix of the periodic cell, and $\mathbf{k} \in \mathbb{Z}^3$ is the periodic cell offset vector. To preserve directional information, hidden states $\mathbf{h}_i^{(t)}$ of atom i at layer t are organized as a direct sum of $SO(3)$ irreducible representations: $\mathbf{h}_i^{(t)} = \bigoplus_{l=0}^{l_{\max}} \mathbf{f}_i^{(l,t)}$, $\mathbf{f}_i^{(l,t)} \in \mathbb{R}^{C \times (2l+1)}$, where $l_{\max}$ denotes the maximum irreducible representation degree, l is the order of the $SO(3)$ irreducible representation, C is the number of feature channels for each representation degree, and $\mathbf{f}_i^{(l,t)}$ is the feature component of degree l at layer t . The intuition behind this design is to explicitly separate features according to their angular variation patterns. Each degree l captures a distinct type of directional dependence (e.g., $l = 1$ for dipole-like, $l = 2$ for quadrupole-like), so the network can naturally distinguish different orientational interactions in the local chemical environment while remaining strictly rotation-equivariant. These representations are updated via equivariant attention blocks:

$$\mathbf{h}_i^{(t+1)} = \text{SLN} \left(\mathbf{h}_i^{(t)} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(t)} \mathbf{V}^{(t)} \left(\tau(\mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(l)}) \right) \right), \quad (1)$$

where $\mathcal{N}(i)$ represents the set of neighbor atoms of atom i , $\alpha_{ij}^{(t)} \in \mathbb{R}$ is the scalar attention coefficient between atom i and atom j at layer t , $\mathbf{e}_{ij}^{(l)} \in \mathbb{R}^{(2l+1)}$ denotes the $SO(3)$ -equivariant edge features derived from relative atomic positions, $\tau(\cdot)$ denotes the $SO(3)$ -equivariant tensor product operation, $\mathbf{V}^{(t)} : \mathbb{R}^{C \times (2l+1)} \rightarrow \mathbb{R}^{C \times (2l+1)}$ is the learnable linear projection at layer t , and SLN denotes separable layer normalization for stable training of equivariant features. The tensor product $\tau(\cdot)$ and linear projection $\mathbf{V}^{(t)}$ act on each $SO(3)$ irreducible component indexed by l separately, ensuring that messages exchanged between atoms strictly follow the rotation rules of the group.

The final node representations after T EquiformerV2 layers are obtained by $\mathbf{g}_i = \mathbf{h}_i^{(T)}$, and the graph representations are obtained via a pooling operator: $\mathbf{g} = \text{Pooling}(\{\mathbf{g}_i\})$. By construction, $\mathbf{g}$ maintains rotation equivariance, transforming predictably via Wigner-D matrices under any rotation.

Descriptor tokenizer. The textual descriptors $\mathcal{S}$ are processed by a text encoder based on the RoBERTa architecture [53]. Given initial token embeddings $\mathbf{u}^{(0)} = \mathcal{S}$, we use a stack of M Transformer layers containing multi-head self-attention (MHA) and feed-forward networks (FFN) to encode the textual representations. The update rule for the textual token representations at layer m is: $\mathbf{u}^{(m+1)} = \text{FFN}(\text{MHA}(\mathbf{u}^{(m)}))$, where MHA captures long-range dependencies among the chemical descriptors, and FFN is responsible for the non-linear transformations of token features. The final semantic features $\mathbf{u}$ encode the critical expert knowledge for characterizing the adsorption system.

C. Group-Conditioned Optimal Transport

Soft group assignment. Directly matching raw individual samples often yields unstable alignment, as highly scattered feature representations can easily perturb the optimal transport solution. To address this issue, we adopt a soft grouping strategy. Instead of directly matching raw samples, we first map them to a set of group prototypes, which constrains the transport problem and improves its stability. Rather than assigning each sample to a single hard group, we compute membership scores indicating the degree to which each sample belongs to every prototype. This soft assignment preserves uncertainty for samples near group boundaries, thereby enhancing alignment robustness.

Specifically, given the full textual feature matrix $\bar{\mathbf{U}} = [\mathbf{u}_1, \dots, \mathbf{u}_N] \in \mathbb{R}^{d \times N}$, we initialize learnable semantic group prototypes $\mathbf{C}^{\text{text}} = [\mathbf{c}_1^{\text{text}}, \dots, \mathbf{c}_S^{\text{text}}] \in \mathbb{R}^{d \times S}$. Then, we define the textual soft assignment matrix $\mathbf{A} \in \mathbb{R}^{N \times S}$ with row-wise softmax normalization: $\mathbf{A}_{ki} = \frac{\exp(\mathbf{u}_k^T \mathbf{c}_i^{\text{text}})}{\sum_{i'=1}^S \exp(\mathbf{u}_k^T \mathbf{c}_{i'}^{\text{text}})}$, where $\mathbf{A}_{ki}$ represents the membership degree of textual sample k belonging to the i -th semantic group. Each row of $\mathbf{A}$ constitutes a valid probability distribution over all group prototypes, and larger entry values indicate higher similarity between the sample embedding and the corresponding group center.

We optimize semantic prototypes by jointly minimizing a reconstruction loss and maximizing an entropy regularization term, which avoids trivial collapsed assignments where all samples fall into a single group:

$$\min_{\mathbf{C}^{\text{text}}} \left\| \mathbf{A} (\mathbf{C}^{\text{text}})^T - \bar{\mathbf{U}}^T \right\|_F^2 - \lambda_h \frac{1}{N} \sum_{k=1}^N H(\mathbf{A}_{k,:}), \quad (2)$$

in which λ_h controls entropy regularization strength, and $H(\cdot)$ stands for Shannon entropy. The reconstruction term drives group centers to capture inherent semantic patterns, while the entropy term preserves assignment diversity and facilitates stable group-level matching. Without the entropy term, the minimization would often collapse all assignments to a single prototype, losing any meaningful grouping.

Similarly, we derive geometric soft assignment based on the geometric feature matrix $\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_N] \in \mathbb{R}^{d \times N}$ and geometric prototypes $\mathbf{C}^{\text{geo}} = [\mathbf{c}_1^{\text{geo}}, \dots, \mathbf{c}_S^{\text{geo}}] \in \mathbb{R}^{d \times S}$. The geometric assignment matrix $\mathbf{B} \in \mathbb{R}^{N \times S}$ is formulated as: $\mathbf{B}_{lj} = \frac{\exp(\mathbf{g}_l^T \mathbf{c}_j^{\text{geo}})}{\sum_{j'=1}^S \exp(\mathbf{g}_l^T \mathbf{c}_{j'}^{\text{geo}})}$, where $\mathbf{B}_{lj}$ quantifies the membership of geometric sample l to the j -th geometric group.

Soft group-based modality alignment. With soft assignment scores obtained, we construct grouped feature sets $\{\mathbf{U}_i \in \mathbb{R}^{d \times N}\}_{i=1}^S$ and $\{\mathbf{G}_j \in \mathbb{R}^{d \times N}\}_{j=1}^S$. Each $\mathbf{U}_i$ stacks the features of all textual samples as columns (each column is a d -dimensional embedding vector), and each $\mathbf{G}_j$ contains the full set of geometric samples arranged in the same column-wise fashion. Different groups are distinguished only by their soft membership weights, not by hard partitions. All groups share the total sample number N , as every sample contributes to all groups via continuous membership scores.

To turn the membership scores into probability distributions that can be compared via optimal transport, we first compute group-level empirical probability measures μ_i and ν_j for textual and geometric groups, respectively. The idea is to treat each

group as a discrete distribution over the sample points, where the probability mass of a point is proportional to how strongly it belongs to that group:

$$\mu_i := \sum_{k=1}^N \mathbf{a}_k^{(i)} \delta_{\mathbf{U}_i(k)}, \quad \nu_j := \sum_{l=1}^N \mathbf{b}_l^{(j)} \delta_{\mathbf{G}_j(l)}, \quad (3)$$

where $\mathbf{U}_i(k)$ and $\mathbf{G}_j(l)$ denote the k -th column of $\mathbf{U}_i$ and the l -th column of $\mathbf{G}_j$, i.e., the feature vectors of the k -th textual sample and the l -th geometric sample, respectively. The normalized weights are obtained from the soft assignment matrices $\mathbf{A}$ and $\mathbf{B}$: $\mathbf{a}_k^{(i)} = \frac{\mathbf{A}_{ki}}{\sum_{k'} \mathbf{A}_{k'i}}$, $\mathbf{b}_l^{(j)} = \frac{\mathbf{B}_{lj}}{\sum_{l'} \mathbf{B}_{l'j}}$, and $\delta_{\mathbf{x}}$ denotes the Dirac measure concentrated at point $\mathbf{x}$. The normalization ensures that $\sum_k \mathbf{a}_k^{(i)} = 1$ and $\sum_l \mathbf{b}_l^{(j)} = 1$, so that each μ_i and ν_j is a valid probability distribution. Intuitively, the weight $\mathbf{a}_k^{(i)}$ tells us how representative sample k is for group i , and points with low membership contribute very little mass.

To compare a textual group μ_i with a geometric group ν_j , we use the squared 2-Wasserstein distance, which measures the minimal total squared Euclidean cost of transporting the mass of μ_i onto ν_j :

$$W_2^2(\mu_i, \nu_j) := \min_{\mathbf{Q} \in \mathcal{U}(\mathbf{a}^{(i)}, \mathbf{b}^{(j)})} \sum_{k=1}^N \sum_{l=1}^N \mathbf{Q}(k, l) \|\mathbf{U}_i(k) - \mathbf{G}_j(l)\|_2^2, \quad (4)$$

where the transport matrix $\mathbf{Q} \in \mathbb{R}^{N \times N}$ specifies how much mass moves from textual point k to geometric point l . The admissible transport plans must respect the marginal masses exactly:

$$\sum_l \mathbf{Q}(k, l) = \mathbf{a}_k^{(i)}, \quad \sum_k \mathbf{Q}(k, l) = \mathbf{b}_l^{(j)}, \quad \mathbf{Q} \geq \mathbf{0}.$$

We formalize this set, the transport polytope, as

$$\mathcal{U}(\mathbf{a}, \mathbf{b}) := \{\mathbf{Q} \in \mathbb{R}_+^{N \times N} : \mathbf{Q} \mathbf{1}_N = \mathbf{a}, \mathbf{Q}^\top \mathbf{1}_N = \mathbf{b}\}, \quad (5)$$

with $\mathbf{1}_N$ the all-one vector of length N . In formulation (4), $\|\cdot\|_2$ is the standard Euclidean norm on $\mathbb{R}^d$.

Unsupervised alignment is challenging because the optimization landscape contains many local minima; imposing additional structure helps prune them away. Our key insight is that a hierarchical structure decomposes a complicated optimization surface into simpler ones that are far less prone to local minima. Instead of aligning the full, ungrouped distributions at the sample level in one shot, we first match coarser group-level summaries and then refine the within-group pointwise alignments. This two-stage matching dramatically constrains the search space and removes many spurious local minima.

We thus formulate a hierarchical Wasserstein approach that jointly learns an alignment transformation T and the group-correspondence matrix $\mathbf{P}$:

$$\min_{\mathbf{P} \in \mathcal{B}_S, T \in \mathcal{T}} \sum_{i=1}^S \sum_{j=1}^S \mathbf{P}_{ij} W_2^2(T(\mu_i), \nu_j), \quad (6)$$

where $\mathbf{P} \in \mathbb{R}^{S \times S}$ encodes the strength of correspondences between soft groups. A large $\mathbf{P}_{ij}$ indicates a strong link between textual group i and geometric group j ; a small value indicates a weak or absent link. The set $\mathcal{B}_S := \mathcal{U}(\mathbf{1}_S/S, \mathbf{1}_S/S)$

is the S -th Birkhoff polytope (doubly stochastic matrices). This formulation is a nested optimal transport problem: the outer level resolves soft group-correspondences through $\mathbf{P}$, while the inner level resolves point-wise correspondences inside each group pair through the Wasserstein distance. By separating global group matching from fine-grained point matching, the optimization becomes more stable and interpretable: $\mathbf{P}$ learns a soft matching between groups of similar structures, while the inner transport model the detailed geometric transformations that map one modality onto the other.

D. Optimization for Group-Conditioned Optimal Transport

Alignment over the Stiefel manifold. Recall that in the hierarchical formulation (6), we need to learn an alignment transformation T that maps each textual group distribution μ_i into the geometric feature space. We now specify the form of T . Assuming that groups approximately lie on low-dimensional subspaces and that the principal angles between these subspaces are “well preserved” across the two modalities (we make this precise in Theorem 2), it suffices to use an isometric transformation. Intuitively, if the internal geometric fabric of a group — the mutual distances and angles among its points — is similar in both text and geometry, then a global rotation or reflection can align the two views without stretching or squeezing the features. Such transformations are exactly represented by orthogonal matrices.

We therefore restrict T to the Stiefel manifold $\mathcal{V}_{d,d}$, the set of all $d \times d$ orthogonal matrices:

$$\mathcal{V}_{d,d} := \{\mathbf{R} \in \mathbb{R}^{d \times d} : \mathbf{R}^\top \mathbf{R} = \mathbf{I}\}. \quad (7)$$

For any $\mathbf{R} \in \mathcal{V}_{d,d}$, applying $\mathbf{R}$ to a textual feature vector simply rotates or reflects it, which preserves Euclidean distances and angles. Consequently, the alignment does not arbitrarily distort the semantic space; it only brings it into correspondence with the geometric space through a rigid motion.

With this choice, the general alignment $T(\mu_i)$ becomes the transformed distribution obtained by replacing each support point $\mathbf{U}_i(k)$ with $\mathbf{R}\mathbf{U}_i(k)$, while keeping the same probability weights $\mathbf{a}^{(i)}$. Plugging this into the hierarchical optimal transport problem (6) yields the concrete optimization:

$$\min_{\mathbf{P}, \mathbf{R}, \{\mathbf{Q}_{ij}\}} \sum_{i,j} \mathbf{P}_{ij} C_{ij}(\mathbf{R}, \mathbf{Q}_{ij}) \quad \text{s.t.} \quad \begin{aligned} \mathbf{P} &\in \mathcal{B}_S, \mathbf{R} \in \mathcal{V}_{d,d}, \\ \mathbf{Q}_{ij} &\in \mathcal{U}(\mathbf{a}^{(i)}, \mathbf{b}^{(j)}), \end{aligned} \quad (8)$$

where the pairwise group cost is defined as

$$C_{ij}(\mathbf{R}, \mathbf{Q}_{ij}) := \sum_{k,l} \mathbf{Q}_{ij}(k, l) \|\mathbf{R}\mathbf{U}_i(k) - \mathbf{G}_j(l)\|_2^2. \quad (9)$$

Here C_{ij} measures the squared 2-Wasserstein distance between the rotated textual group i and the geometric group j , using the transport plan $\mathbf{Q}_{ij}$. Because $\mathbf{R}$ is orthogonal, the norm of any transformed vector remains unchanged, so the intrinsic structure of the textual features is preserved during alignment.

Directly solving formulation (8) is computationally demanding because the Wasserstein distances involve constrained optimization over $\mathbf{Q}_{ij}$, and the entire problem is non-convex. To overcome this, we introduce entropic regularization on

all transport couplings. This replaces the exact Wasserstein distances with their Sinkhorn counterparts, which can be computed efficiently via iterative matrix scaling and lead to smoother optimization landscapes with better sample complexity. Omitting the constraints for brevity, the final objective becomes

$$\min_{\mathbf{P}, \mathbf{R}, \{\mathbf{Q}_{ij}\}} \sum_{i,j} (\mathbf{P}_{ij} C_{ij}(\mathbf{R}, \mathbf{Q}_{ij}) + \lambda_2 H(\mathbf{Q}_{ij})) + \lambda_1 H(\mathbf{P}), \quad (10)$$

where $\lambda_1, \lambda_2 > 0$ are the entropic regularization parameters, and $H(\cdot)$ denotes the Shannon entropy, e.g., $H(\mathbf{P}) := \sum_{i,j} \mathbf{P}_{ij} \log \mathbf{P}_{ij}$. The entropy terms penalize overly deterministic transport plans, encouraging the model to spread mass more uniformly. As $\lambda_1, \lambda_2 \rightarrow 0$, the solution of the formulation (10) approaches that of the original unregularized problem (8); larger values yield smoother, more explorative plans that are easier to optimize but introduce additional bias. In practice, the parameters λ_1, λ_2 are tuned to balance computational tractability and alignment accuracy.

ADMM approach. Problem (10) is difficult to solve directly because it is non-convex: the variables $\mathbf{P}$, $\mathbf{R}$, and $\{\mathbf{Q}_{ij}\}$ are multiplied together in the objective, and $\mathbf{R}$ is constrained to lie on the Stiefel manifold $\mathcal{V}_{d,d}$, which is not a convex set. Although the alternating direction method of multipliers (ADMM) was originally developed as a convergent solver for convex problems [54], [55], it has been successfully applied to many non-convex settings [56] where the problem structure allows efficient local updates. A key structural feature of our problem is that the objective separates naturally across group pairs (i, j) : each term $\mathbf{P}_{ij} C_{ij}(\mathbf{R}, \mathbf{Q}_{ij}) + \lambda_2 H(\mathbf{Q}_{ij})$ involves only the local coupling $\mathbf{Q}_{ij}$ and the global rotation $\mathbf{R}$, with no direct coupling between different (i, j) blocks besides the shared $\mathbf{R}$. This immediately suggests a divide-and-conquer strategy: we can assign each group pair its own local rotation estimate $\mathbf{R}_{ij}$, allow them to solve for an optimal $\mathbf{R}_{ij}$ independently and in parallel, and then enforce consensus among all local estimates to obtain a single global rotation $\tilde{\mathbf{R}}$.

To implement this idea, we introduce auxiliary variables $\mathbf{R}_{ij}$ for each group pair and explicitly require them to agree with a common global variable $\tilde{\mathbf{R}}$. The problem then becomes

$$\begin{aligned} \min_{\mathbf{P}, \tilde{\mathbf{R}}, \{\mathbf{R}_{ij}, \mathbf{Q}_{ij}\}} \sum_{i,j} (\mathbf{P}_{ij} C_{ij}(\mathbf{R}_{ij}, \mathbf{Q}_{ij}) + \lambda_2 H(\mathbf{Q}_{ij})) + \lambda_1 H(\mathbf{P}) \\ \text{s.t. } \mathbf{R}_{ij} = \tilde{\mathbf{R}}, \forall i, j, \end{aligned} \quad (11)$$

where the set constraints on $\mathbf{P}$, $\mathbf{R}_{ij}$ (Stiefel), and $\mathbf{Q}_{ij}$ are omitted for brevity. The equality constraints $\mathbf{R}_{ij} = \tilde{\mathbf{R}}$ are the glue that forces all local rotations to ultimately agree. We then form the scaled augmented Lagrangian to handle these constraints in a numerically stable fashion:

$$\begin{aligned} \mathcal{L}_{\text{map}} = \sum_{i,j} (\mathbf{P}_{ij} C_{ij}(\mathbf{R}_{ij}, \mathbf{Q}_{ij}) + \frac{\mu}{2} \|\mathbf{R}_{ij} - \tilde{\mathbf{R}}\|_F^2 \\ + \langle \mu \mathbf{\Lambda}_{ij}, \mathbf{R}_{ij} - \tilde{\mathbf{R}} \rangle + \lambda_2 H(\mathbf{Q}_{ij})) + \lambda_1 H(\mathbf{P}), \end{aligned} \quad (12)$$

where $\mu > 0$ is the ADMM penalty parameter that controls the strength of consensus enforcement, and $\{\mathbf{\Lambda}_{ij}\}$ are the scaled

Lagrange multiplier matrices associated with each constraint $\mathbf{R}_{ij} = \tilde{\mathbf{R}}$. Full details of the ADMM algorithm are presented in the Supplementary Material.

E. Prediction Head

From text to geometry-aligned features. The group-conditioned optimal transport module not only aligns the distributions of the two modalities, but also produces a concrete mapping that transports textual features into the geometric latent space. This mapping is given by the learned orthogonal Stiefel transformation $\mathbf{R} \in \mathcal{V}_{d,d}$, which we obtained by solving problem (10). Because $\mathbf{R}$ is orthogonal, it preserves norms and inner products; applying it to a textual embedding rotates the vector into the geometric coordinate system without distorting its structure. For any textual sample i with semantic embedding $\mathbf{u}_i$, the aligned geometric-aware representation is simply $\hat{\mathbf{g}}_i = \mathbf{R} \mathbf{u}_i$. In the training stage, this transformation is applied after the GCOT alignment has converged; during fine-tuning it remains fixed.

Learning objectives. With the aligned embeddings $\hat{\mathbf{g}}_i$ in hand, the adsorption energy prediction reduces to a standard regression problem. A lightweight prediction head $h(\cdot)$, typically a small multi-layer perceptron, maps each $\hat{\mathbf{g}}_i$ to a scalar energy estimate. To make training numerically stable, we first standardize the ground-truth adsorption energies using z-score normalization: $\bar{y}_i = (y_i - \mu)/\sigma$, where μ and σ are the mean and standard deviation computed over the entire training set. The head therefore predicts normalized values $\hat{y}_i = h(\hat{\mathbf{g}}_i)$, and we train it with the mean absolute error (MAE) loss, which is robust to outliers and directly measures average prediction accuracy in the normalized space: $\mathcal{L}_{\text{energy}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} |\hat{y}_i - \bar{y}_i|$, where $\mathcal{B}$ denotes the current mini-batch. The full training objective combines the alignment loss and the energy regression loss: $\mathcal{L} = \mathcal{L}_{\text{map}} + \lambda \mathcal{L}_{\text{energy}}$, where $\mathcal{L}_{\text{map}}$ is the optimal transport loss from the GCOT module.

To prevent overfitting on the typically limited adsorption data and to retain the powerful geometric priors already captured by the pre-trained encoder and the GCOT alignment, we adopt a selective fine-tuning strategy: The geometric encoder and all GCOT alignment modules, including the Stiefel transformation $\mathbf{R}$, are kept frozen. Only the semantic encoder and the energy regression head $h(\cdot)$ are updated. This design ensures that the geometry-encoding branch and the hard-won cross-modal mapping remain intact, while the text processing and the final predictor can still adapt to the specifics of the adsorption energy task.

F. Geometry-Free Inference

Once training is complete, the model can predict adsorption energies from pure textual input, without ever seeing the corresponding 3D atomic structure. For an unseen textual descriptor with embedding $\mathbf{u}^*$, we use exactly the same frozen Stiefel transformation $\mathbf{R}$ to project it into the geometric latent space: $\hat{\mathbf{g}}^* = \mathbf{R} \mathbf{u}^*$. The aligned feature is then fed to the energy regression head to obtain the prediction. The entire pipeline thus accepts a textual description and returns an adsorption energy, realizing efficient, geometry-free prediction that inherits

the structural knowledge distilled from the crystal data through the GCOT alignment.

G. Theoretical Analysis

This section theoretically analyzes the alignability and alignment quality of the proposed GCOT framework. We derive sufficient conditions for group-level alignment feasibility and establish rigorous perturbation bounds for the optimal alignment transformation based on the GCOT optimization formulation (8). To facilitate theoretical analysis, we adopt standard simplifying assumptions: (i) all groups contain an equal number of sampled points n ; (ii) the ground-truth group correspondence matrix is set to $\mathbf{P}^* = \mathbf{I}_S/S$, i.e., diagonal containing $1/S$. Notably, this analysis can be trivially extended to unbalanced group sizes without loss of generality. Detailed mathematical proofs are provided in the Supplementary Material.

The following result is a criterion that, if met, ensures the existence of a global minimizer of the group-correspondence $\mathbf{P}^*$. This criterion requires that matched groups must be closer in Wasserstein distance than mismatched groups, according to a threshold determined by Wasserstein's sample complexity (i.e., an asymptotic rate dependent on the groups' sample sizes and intrinsic dimensions). Since these sample complexity results are based on the Wasserstein distance, we expect a less stringent criterion when using the Sinkhorn distance in (10) due to superior sample complexity [57].

Theorem 1 (Correspondence Disambiguity Criterion). *Let all groups be strictly low-rank, where the intrinsic dimension of the i -th group is $d_{x,i}$ and $d_{x,i}, d_{y,j} > 4$ for all group indices $i, j \in [1, S]$. Define the minimal group-wise transport cost*

$$C_{ij}^* := \min_{\mathbf{R} \in \mathcal{V}_{d,d}, \mathbf{Q}_{ij} \in \mathcal{U}(\mathbf{a}^{(i)}, \mathbf{b}^{(j)})} C_{ij}(\mathbf{R}, \mathbf{Q}_{ij}).$$

The GCOT optimization in (8) yields the unique global optimal group correspondence $\mathbf{P}^ = \mathbf{I}_S/S$ with probability at least $1 - \epsilon$ if for all $i \neq j$:*

$$C_{ij}^* + C_{ji}^* - C_{ii}^* - C_{jj}^* > B_{x,i}(\epsilon) + B_{y,i}(\epsilon) + B_{x,j}(\epsilon) + B_{y,j}(\epsilon),$$

where the finite-sample error bound is formulated as $B_{z,k}(\epsilon) := c_{z,k} n^{-\frac{2}{d_{z,k}}} + \sqrt{\frac{\log(1/\epsilon)}{2n}}$ and $c_{z,k} = 729 \left(2 + \frac{1}{2 \cdot 3^{d_{z,k}/2 - 1}}\right)$.

Proof sketch. The proof consists of two core parts. First, we derive the perturbation condition for unique optimal transport matching on the Birkhoff polytope, requiring a positive cost margin $C_{ij} + C_{ji} - C_{ii} - C_{jj} > 0$ to guarantee exact diagonal group correspondence. Second, we incorporate finite-sample concentration bounds of 2-Wasserstein distance [58] to correct the ideal condition for practical empirical group distributions, resulting in the final robust disambiguity threshold.

Now, even if we know the global correspondence $\mathbf{P}^*$, we still do not have the full picture about the alignment's quality. For example, all matching groups may have very similar covariances, but principal angles between the groups are "distorted" across the datasets. Our next theorem gives us an upper bound on the alignment error (for unitary transformations), and makes precise the notion of global structure distortion.

Theorem 2 (Group-Based Alignment Perturbation Bound). *Given paired group feature matrices $\{\mathbf{U}_i, \mathbf{G}_j \in \mathbb{R}^{d \times n}\}_{i,j=1}^S$ with optimal point-wise transport couplings $\{\mathbf{Q}_{ii} \in \mathcal{U}(\mathbf{a}^{(i)}, \mathbf{b}^{(i)})\}_{i=1}^S$, we define feature matrices*

$$\mathbf{U}_{agg} = [\mathbf{U}_1 \mathbf{Q}_{11}, \mathbf{U}_2 \mathbf{Q}_{22}, \dots, \mathbf{U}_S \mathbf{Q}_{SS}],$$

$$\mathbf{G}_{agg} = [\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_S]$$

Let the global structural distortion error be $\epsilon := \|\mathbf{G}_{agg}^\top \mathbf{G}_{agg} - \mathbf{U}_{agg}^\top \mathbf{U}_{agg}\|_F$. Under the condition that Theorem 1 holds, $\mathbf{U}_{agg}$ is full row-rank, and $\epsilon \|\mathbf{U}_{agg}^\dagger\| \leq \frac{1}{\sqrt{2}} (\|\mathbf{U}_{agg}\| \|\mathbf{U}_{agg}^\dagger\|)^{-1/2}$, the optimal GCOT objective is bounded as:

$$\min_{\substack{\mathbf{P} \in \mathcal{B}_S, \\ \mathbf{R} \in \mathcal{V}_{d,d}}} \sum_{i,j} \mathbf{P}_{ij} C_{ij}(\mathbf{R}, \mathbf{Q}_{ij}) \leq (\|\mathbf{U}_{agg}\| \|\mathbf{U}_{agg}^\dagger\| + 2)^2 \|\mathbf{U}_{agg}^\dagger\|^2 \epsilon^2 + \mathcal{D},$$

where $\mathcal{D} = \sum_{i=1}^S \text{Tr}(\mathbf{U}_i (\frac{1}{n} \mathbf{Q}_{ii} \mathbf{Q}_{ii}^\top - \mathbf{I}) \mathbf{U}_i^\top + (\frac{1}{n} - 1) \mathbf{G}_i \mathbf{G}_i^\top)$ denotes a data-dependent constant term.

Proof sketch. We extend the classical Procrustes perturbation analysis on Frobenius-norm alignment objectives [59] to our hierarchical Wasserstein-based GCOT formulation, deriving the final alignment error bound (see Supplementary Material for full proof). The distortion term ϵ dominates the alignment error bound and explicitly quantifies cross-domain structural misalignment caused by covariate shift, imperfect soft group, or domain heterogeneity. To further interpret this term, we decompose the global distortion ϵ into pairwise group-level distortion ϵ_{ij} . For paired groups i, j , we perform singular value decomposition on aligned group features: $\mathbf{U}_i \mathbf{Q}_{ii} = \mathbf{J}_i \Sigma_{u,i} \mathbf{V}^\top$, $\mathbf{G}_j = \mathbf{K}_j \Sigma_{g,j} \mathbf{V}^\top$. The group-wise distortion is defined as:

$$\epsilon_{ij} := \|\mathbf{G}_i^\top \mathbf{G}_j - \mathbf{Q}_{ij}^\top \mathbf{U}_i^\top \mathbf{U}_j \mathbf{Q}_{jj}\|_F$$

$$= \|\Sigma_{g,i} \mathbf{K}_i^\top \mathbf{K}_j \Sigma_{g,j} - \Sigma_{u,i} \mathbf{J}_i^\top \mathbf{J}_j \Sigma_{u,j}\|_F,$$

which decomposes structural distortion into two interpretable components: (1) angular shift, originating from inconsistent principal angles ($\mathbf{J}_i^\top \mathbf{J}_j$ and $\mathbf{K}_i^\top \mathbf{K}_j$) between cross-domain subspaces; (2) spectral shift, originating from mismatched singular value spectra ($\Sigma_{g,\cdot}$ and $\Sigma_{u,\cdot}$) of paired groups.

Finally, we show that the subspace configuration of a dataset's groups can also affect alignment. Pretend for a moment that external alignment information were present to aid in the disambiguation between two groups. The following lemma tells us when such information is useless (Proof in Supplemental Materials).

Lemma 1 (Uninformative Alignment Condition). *Given paired groups $\mathbf{U}_i, \mathbf{G}_j \in \mathbb{R}^{d \times n}$ and point-wise transport matrix $\mathbf{Q}_{ij} \in \mathcal{U}(\mathbf{a}^{(i)}, \mathbf{b}^{(j)})$, let $\tilde{\mathbf{J}}, \tilde{\mathbf{K}} \in \mathbb{R}^{d \times r}$ denote the singular vectors corresponding to non-zero singular values of $\mathbf{G}_j \mathbf{Q}_{ij}^\top \mathbf{U}_i^\top$. Define the constrained orthogonal transformation set $\mathcal{T}(\mathbf{J}', \mathbf{K}') := \{\mathbf{R} \in \mathcal{V}_{d,d} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}, \mathbf{R} \mathbf{K}' = \mathbf{J}'\}$ that enforces alignment with prior geometric directions. For any valid $\mathbf{J}', \mathbf{K}'$, we have:*

$$\min_{\mathbf{R} \in \mathcal{T}(\mathbf{J}', \mathbf{K}')} C_{ij}(\mathbf{R}) = \min_{\mathbf{R} \in \mathcal{V}_{d,d}} C_{ij}(\mathbf{R}),$$

with equality if and only if the principal angles of prior constraints match the intrinsic subspace angles of group pairs $\langle \tilde{\mathbf{J}}, \mathbf{J}' \rangle = \langle \tilde{\mathbf{K}}, \mathbf{K}' \rangle$.

This lemma reveals two key insights for our geometry-free inference pipeline. First, datasets with equally spaced or mutually orthogonal subspaces belong to the worst-case alignment scenario, where external geometric constraints provide no additional alignment information. Second, high-dimensional feature spaces tend to produce degenerate orthogonal subspaces, making cross-domain alignment more challenging.

IV. RESULTS AND DISCUSSION

A. Benchmark Datasets

The OC dataset marks a pivotal advancement in computational catalysis, comprising more than 1.2 million DFT structural relaxation samples that span 55 elemental species and 82 adsorbate types [60]. Covering abundant nitrogen, carbon and oxygen reaction chemistries over varied inorganic material surfaces. It facilitates efficient data-driven catalyst exploration targeting solar fuel production and renewable energy storage applications.

OC20 IS2RE. In this work, we focus on the Initial Structure to Relaxed Energy (IS2RE) task in OC20. Given an unrelaxed initial adsorption structure, the goal is to predict the final relaxed adsorption energy without carrying out DFT relaxation. We evaluate the model under three training settings: Train-10k, Train-100k, and the full-scale 460k setting. These settings allow us to examine model behavior in both low-data and large-scale regimes. Since the DFT relaxation trajectories for the test data set are not publicly accessible, our evaluation and comparison of model performance are centered on the validation data set. The database provides four official validation splits to systematically evaluate model generalization: The **ID** split contains adsorbates and catalysts seen during training but uses new geometries. The **OOD_{ads}** split contains unseen adsorbates on known catalysts, while the **OOD_{cat}** split contains known adsorbates on unseen catalyst compositions. The **OOD_{both}** split contains both unseen adsorbates and unseen catalyst materials. For evaluation, we randomly sample 2,500 unique configurations from each validation split and combine them into a **Total** set of 10,000 configurations, which reports predictive accuracy across these distribution shifts.

MamunHighT2019. To complement the Open Catalyst Project benchmark, we additionally evaluate on the CatBench-formatted MamunHighT2019 adsorption dataset [61], which is curated from Catalysis-Hub and provides DFT-referenced adsorption energies for heterogeneous catalysis. Each configuration contains an adsorbate-surface structure and a reference adsorption energy. The processed dataset contains 45,130 unique configurations, spanning 15 adsorbates across 1,916 bulk materials. We construct a reproducible train and validation protocol with one training split and four validation subsets, namely **ID**, **OOD_{ads}**, **OOD_{cat}**, and **OOD_{both}**, defined by adsorbate and catalyst-composition overlap with the training set.

OC20Dense. Beyond energy prediction on OC20 IS2RE and MamunHighT2019, we also evaluate high-throughput

adsorption configuration screening on OC20Dense. Unlike standard IS2RE, where each adsorption structure is treated as an independent regression example, OC20Dense provides multiple candidate adsorption configurations for the same adsorbate-catalyst system. In this benchmark, candidate configurations are ranked by predicted relaxed adsorption energy, and the lowest-energy configuration selected by the model is compared with the DFT global minimum. This setting tests whether a model can identify the most stable adsorption configuration from a dense candidate pool, a central requirement in practical catalyst screening.

OC20Dense complements the OC20 IS2RE and MamunHighT2019 regression benchmarks by moving the evaluation from energy prediction to configuration selection. It also measures ranking quality and the energy prediction error caused by selecting a suboptimal adsorption configuration. Together, these benchmarks assess whether relaxation-free adsorption energy predictors can support both accurate energy estimation and efficient high-throughput configuration screening.

Baselines. We compare *Taco* with three categories of baselines: (1) textual representation-based methods, including Graphormer [62] and CatBERTa [18]; (2) geometry-based methods, including CGCNN [25], DimeNet++ [29], EquiformerV2 [32], and Graphormer [62]; and (3) a multimodal baseline, GAP [38]. Among these, CGCNN and DimeNet++ utilize officially available IS2RE results or pre-trained checkpoints. For Graphormer, CatBERTa, GAP, and EquiformerV2, since pre-trained weights for the IS2RE task have not been released, we trained them using publicly available source code. Detailed hyperparameter settings are provided in Supplemental Materials.

B. Text Descriptors-based Relaxed Energy Prediction

Prediction accuracy. We evaluate the proposed framework on the OC20 IS2RE task, where the geometric tokens obtained from the text-geometry projector (Figure 1(b)) are fed into an energy prediction head to directly predict the final relaxed adsorption energy from an unrelaxed initial state. We adopt three distinct training scales, including 10k, 100k, and 460k samples. Prediction errors are quantified in terms of mean absolute error over the full dataset, alongside in-domain and out-of-domain subsets covering **OOD_{ads}**, **OOD_{cat}** and **OOD_{both}**.

As summarized in Table I, *Taco* yields competitive and even advantageous results across diverse evaluation scenarios, and manifests prominent superiority especially under low-data conditions. When trained on 10k samples, *Taco* delivers the minimal overall MAE of 0.769 ± 0.009 , and surpasses all baseline approaches on both in-domain and **OOD_{cat}** subsets. It also attains the optimal prediction accuracy with MAE reaching 0.776 ± 0.006 on the **OOD_{both}** partition at this data scale. Under the 100k training configuration, our method still ranks first in overall performance with MAE calculated as 0.688 ± 0.008 , and maintains leading accuracy across in-domain, **OOD_{ads}** and **OOD_{cat}** test cases. For the complete 460k training collection, EquiformerV2 presents the best comprehensive performance at 0.576 ± 0.007 . Even so, *Taco* still acquires superior results of 0.524 ± 0.006 for in-domain samples and 0.505 ± 0.007 for the **OOD_{cat}** subset.

TABLE I: MAE for energy prediction across various models on the OC20 IS2RE dataset. The highest result is **bold**. All models are trained and tested under the same configurations.

Data Size	Model	Total	ID	OOD_ads	OOD_cat	OOD_both
10k	Graphormer	0.975±0.004	0.962±0.003	0.972±0.002	0.977±0.011	0.990±0.003
10k	CatBERTa	0.853±0.009	0.770±0.011	0.996±0.003	0.713±0.013	0.933±0.011
10k	GAP	0.838±0.008	0.830±0.009	0.841±0.004	0.843±0.007	0.839±0.013
10k	CGCNN	0.978±0.012	1.045±0.026	0.975±0.014	1.028±0.005	0.862±0.004
10k	DimeNet++	0.884±0.010	0.949±0.016	0.890±0.014	0.888±0.007	0.808±0.004
10k	EquiformerV2	0.776±0.011	0.783±0.013	0.797±0.017	0.734±0.009	0.776±0.006
10k	Taco	0.769±0.009	0.699±0.018	0.893±0.032	0.661±0.016	0.847±0.007
100k	Graphormer	0.708±0.009	0.638±0.007	0.793±0.012	0.650±0.006	0.749±0.011
100k	CatBERTa	0.792±0.011	0.678±0.012	0.944±0.006	0.647±0.016	0.897±0.008
100k	GAP	0.778±0.009	0.699±0.004	0.893±0.008	0.665±0.014	0.856±0.012
100k	CGCNN	0.874±0.013	0.988±0.017	0.824±0.011	0.925±0.003	0.759±0.002
100k	DimeNet++	0.769±0.009	0.812±0.013	0.788±0.011	0.759±0.006	0.718±0.005
100k	EquiformerV2	0.692±0.007	0.670±0.009	0.781±0.011	0.614±0.008	0.703±0.003
100k	Taco	0.688±0.008	0.637±0.012	0.776±0.028	0.597±0.011	0.757±0.005
460k	Graphormer	0.675±0.008	0.612±0.007	0.754±0.008	0.614±0.003	0.720±0.006
460k	CatBERTa	0.755±0.006	0.643±0.004	0.905±0.013	0.601±0.008	0.869±0.008
460k	GAP	0.746±0.009	0.671±0.005	0.850±0.004	0.626±0.006	0.838±0.002
460k	CGCNN	0.819±0.007	0.944±0.011	0.783±0.008	0.861±0.006	0.689±0.004
460k	DimeNet++	0.720±0.009	0.747±0.009	0.738±0.006	0.710±0.012	0.685±0.009
460k	EquiformerV2	0.576±0.007	0.589±0.009	0.599±0.002	0.552±0.005	0.562±0.004
460k	Taco	0.668±0.009	0.524±0.006	0.856±0.011	0.505±0.007	0.789±0.010

TABLE II: MAE for energy prediction across various models on the MamunHighT2019 dataset. The highest result is **bold**. All models are trained and tested under the same configurations.

Model	MAE
Graphormer	0.663±0.011
CatBERTa	0.585±0.057
GAP	0.593±0.051
CGCNN	0.702±0.024
DimeNet++	0.462±0.038
EquiformerV2	0.526±0.025
Taco	0.430±0.033

Additional validation is conducted on the MamunHighT2019 dataset, which collects adsorption energy data of metal surfaces measured under varied thermodynamic environments. According to the statistical outcomes listed in Table III, *Taco* obtains the lowest MAE value of 0.430 ± 0.033 , exceeding representative competitors such as DimeNet++ with 0.462 ± 0.038 and EquiformerV2 with 0.526 ± 0.025 . All experimental evidence demonstrates that representations constructed from textual descriptors possess strong applicability for scarce data scenarios and unseen catalyst-related out-of-distribution tasks.

We further adopt the coefficient of determination R^2 to quantify the correlation between predicted adsorption energies and reference values obtained from DFT calculations. Scatter diagrams corresponding to models trained with 10k, 100k and 460k samples are illustrated in Figure 2, where pink points denote in-domain samples and purple points represent data from the OOD_{cat} category. The proposed *Taco* maintains superior R^2 performance on both in-domain and out-of-distribution catalyst samples under all training volumes. When trained on

10k samples, our method achieves an in-domain R^2 of 0.712 and an OOD_{cat} R^2 of 0.732, outperforming geometric-based EquiformerV2 with scores of 0.676 and 0.712, as well as text-only CatBERTa with scores of 0.615 and 0.655. With full-scale 460k training data, our approach attains in-domain and out-of-distribution R^2 values of 0.836 and 0.873 respectively. Such results exceed those yielded by EquiformerV2, DimeNet++, CGCNN and CatBERTa, whose corresponding paired R^2 metrics stand at 0.816/0.855, 0.768/0.818, 0.699/0.804 and 0.735/0.792 in sequence.

Notably, while geometric models such as EquiformerV2 exhibit strong in-domain performance, they lag behind *Taco* on OOD_{cat} data at all scales. This highlights the benefit of integrating structured textual descriptors, which encode chemically meaningful adsorption patterns and enhance generalization to unseen catalytic systems. The consistent improvement across scales and domains confirms that the cross-modal alignment strategy in *Taco* effectively leverages both geometric and semantic information to achieve robust and accurate energy predictions.

Inference cost. Beyond predictive accuracy, computational efficiency constitutes another critical requirement for practical large-scale catalyst screening. While pure geometric models achieve reliable regression performance, their heavy structural modeling inevitably leads to substantial inference overhead. By contrast, our cross-modal enables accurate structure-aware prediction without explicit 3D geometry input during inference, yielding a favorable trade-off between precision and speed. As shown in Figure 3(a), *Taco* achieves a prediction MAE of 0.775 eV, demonstrating competitive accuracy comparable to the state-of-the-art EquiformerV2 (0.773 eV) while delivering drastically improved computational efficiency. Critically, *Taco* completes inference in just 3.23 seconds, outperforming other models

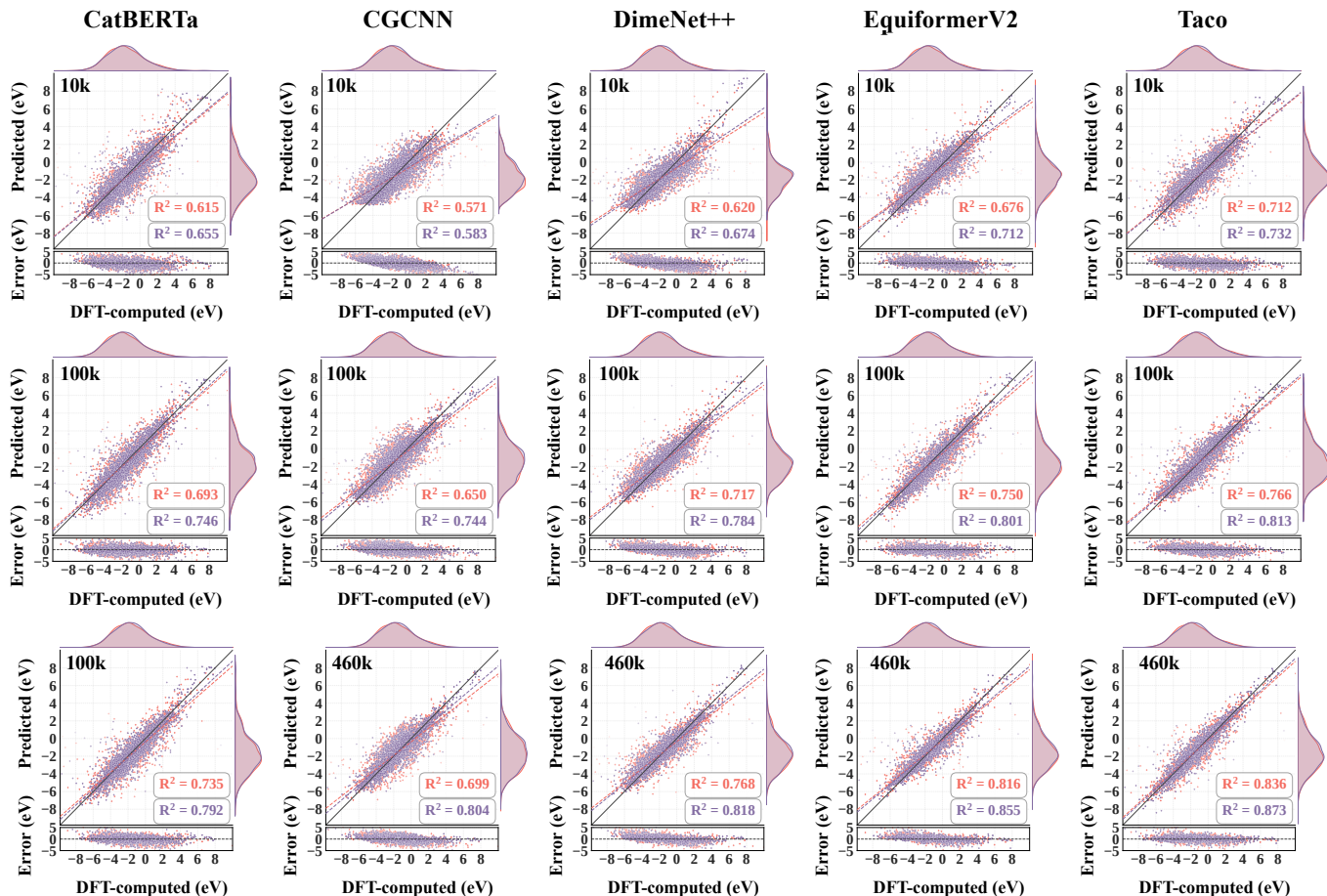


Fig. 2: Coefficient of determination R^2 on the 460k setting. Pink and purple dots denote in-domain and OOD_{cat} data, respectively.

like DimeNet++ and EquiformerV2. *Taco* retains state-of-the-art prediction accuracy while drastically reducing inference overhead, making it uniquely suited for high-throughput catalyst screening applications.

Why does *Taco* outperform purely geometric methods in some cases? Experimental results show that our framework achieves competitive accuracy compared to purely geometric networks. While pure geometric models learn representations directly from 3D atomic structures, the subtle and complex nature of atomic configurations makes them sensitive to irrelevant structural variations. In contrast, structured textual descriptors explicitly encode key physicochemical factors that govern adsorption behavior, thereby providing the model with more reliable inductive priors.

Unlike methods that directly fit raw geometric features of individual samples, our approach performs cross-modal alignment via group-conditioned optimal transport. Once the two modalities are well aligned, the model benefits from both the chemical intuition embedded in text descriptions and the fine-grained precision of geometric features. This leads to stronger predictive performance, particularly in low-data regimes or when encountering unseen catalyst compositions.

The effect of each component. In this part, we conduct ablation studies on the IS2RE task to quantify the individual contributions of *Taco*. We first examine the adaptability of our framework by substituting EquiformerV2 with other mainstream geometric encoders, including SchNet, DimeNet++,

EGNN, and CGCNN, while all other module configurations remain fixed. Figure 3(b) presents the mean absolute error and success rate calculated under the OC20Dense evaluation rule. The success rate quantifies the proportion of tested systems whose predicted configuration energy deviates no more than 0.10 eV from the global minimum energy obtained via DFT relaxation.

EquiformerV2 achieves the optimal comprehensive performance with the smallest prediction error and highest success rate, justifying its selection as the standard geometric encoder for our framework. CGCNN also produces satisfactory results, while DimeNet++, EGNN, and SchNet show relatively weaker prediction capability. Even when equipped with less competitive backbone networks, our integrated framework still maintains reliable predictive performance, which verifies strong flexibility and universal applicability across diverse structural encoders.

We then conduct ablation experiments to validate the complementarity of the two modalities. Two simplified variants are constructed for comparison: one purely relies on EquiformerV2 to conduct energy regression without textual input, and the other removes geometric encoding branches and makes predictions merely based on the chemical descriptions. Quantitative results listed in Table III show that the complete *Taco* framework obtains the minimal MAE of 0.769 eV, outperforming models that adopt only a single information source. Prediction accuracy declines markedly to 1.022 eV for the text-only variant. This phenomenon indicates that semantic information alone cannot

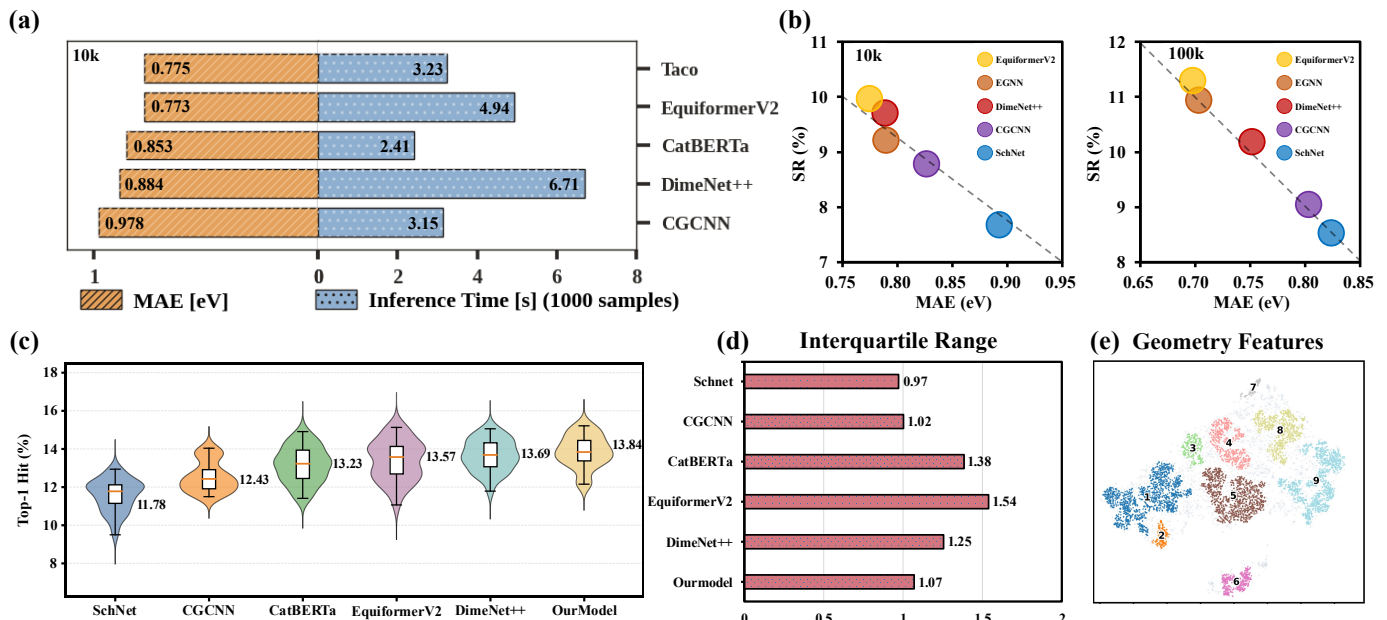


Fig. 3: Model performance evaluation. (a) Prediction error and inference efficiency of various algorithms. (b) Success rate (SR, percentage of predictions with absolute error < 0.1 eV relative to DFT) and mean absolute error (MAE) using different graph encoders on the 10k and 100k datasets. (c) Distribution of Top-1 hit rates across different algorithms in high-throughput computational screening experiments. (d) Interquartile range (IQR) of Top-1 hit rates for various algorithms. (e) t-SNE visualization of geometry features of the OCDense data.

TABLE III: Prediction accuracy and inference speed of *Taco* and its ablated variants on the IS2RE task.

Model / Variant	MAE (eV)	Inference Time (s)
<i>Taco</i> (Full)	0.769	3.23
Geometry-only	0.784	4.89
Text-only	1.022	2.17
w/o Group-level OT	0.793	2.58
w/o Instance-level OT	0.783	2.52

TABLE IV: Relationship between soft group number S , MAE, and OT time cost.

Group Partition S	MAE (eV)	OT Time (s)
1	0.783	1.13
3	0.796	1.19
5	0.785	1.28
7	0.774	1.34
9	0.769	1.38
11	0.770	1.44
13	0.774	1.50
15	0.767	1.57

capture complete three-dimensional structural characteristics, hence failing to fully reflect intrinsic adsorption properties. The geometry-only approach attains comparable accuracy at 0.784 eV, yet its inference time rises to 4.89 seconds, far exceeding the 3.23 seconds consumed by the full model. This efficiency gap demonstrates that compact chemical descriptors offer effective domain priors, bypassing the computational expense of 3D structure processing while retaining essential adsorption characteristics for accuracy.

We further validate the efficacy of our proposed GCOT

module by ablating its two core components: group-level coarse alignment and instance-level fine-grained matching. The variant without group-level alignment presents an increased MAE of 0.793 eV with negligible change in inference speed (2.58 s), indicating that coarse group-level modal correspondence establishes global semantic-geometric alignment and mitigates cross-modal domain misalignment. Meanwhile, removing instance-level fine-grained transport slightly degrades accuracy (MAE = 0.783 eV) without affecting inference efficiency, verifying that instance-level matching refines local cross-modal correspondence and further improves prediction fidelity.

Our proposed GCOT establishes cross-modal associations via group-wise alignment, where groups are defined as data groups. This design is intuitively motivated by the observation that geometric features on OC data can be naturally characterized in a group pattern, as visualized in Fig. 3(e). Based on this finding, we develop the GCOT framework for cross-modal alignment. The performance of GCOT is inherently dependent on the group number configuration. Considering the ambiguous group boundaries inherent to the data, we abandon hard partition strategies and adopt a soft group scheme, which is elaborated in the methodology section. We further conduct a sensitivity analysis study to evaluate the model performance with group numbers ranging from 1 to 15, as reported in Table IV. The experimental results demonstrate that the setting of 1 group yields the highest MAE. When the group number exceeds 7, the MAE value exhibits no significant decline, while the computational overhead of optimal transport calculation increases. Consequently, we finally set the group number to 9. This configuration not only aligns with the visual observations in Figure 3(e), but also achieves a favorable trade-off between prediction accuracy and computational cost.

TABLE V: Adsorption configuration screening results on the OC20Dense benchmark. The best result in each column is shown in **bold**.

Model	Top-1 Hit	Top-2 Hit	Top-3 Hit	SR %	Forward Samples/s
CatBERTa	13.23±1.85	24.74±1.74	36.47±1.72	46.74±0.48	414.85
SchNet	11.96±1.59	20.51±1.21	31.84±1.45	44.08±0.35	155.09
CGCNN	12.51±1.53	23.22±1.42	33.18±1.58	45.46±0.45	317.50
DimeNet++	13.28±1.74	25.59±1.36	36.16±1.59	46.92±0.72	149.13
EquiformerV2	13.40±2.07	26.20±1.51	37.03±1.75	47.25±0.94	202.54
Taco	14.26±1.62	27.34±1.34	38.12±1.61	48.83±0.57	309.20

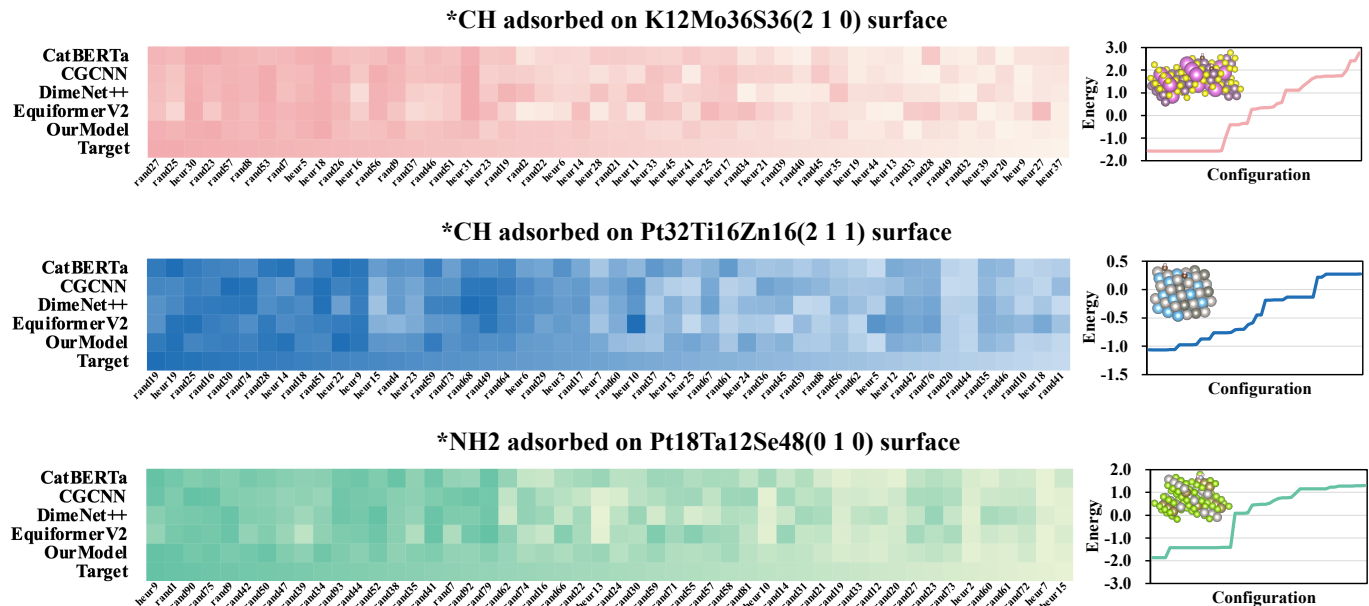


Fig. 4: High-throughput screening validation on three adsorbate–catalyst systems. Left: Systems evaluated: (1) *CH on $K_{12}Mo_{36}S_{36}(2\ 1\ 0)$; (2) *CH on $Pt_{32}Ti_{16}Zn_{16}(2\ 1\ 1)$; (3) *NH₂ on $Pt_{18}Ta_{12}Se_{48}(0\ 1\ 0)$. For each, over 40 initial configurations (heuristic and random) are evaluated with DFT-relaxed energies. Right: Corresponding energy landscapes exhibit three distinct patterns: steep transition, gradual slope, and bimodal distribution, providing a rigorous testbed for model robustness.

C. Text Descriptors-based Fast Configuration Screening

Screening capability. To assess the practical utility of *Taco* in high-throughput computational screening, we evaluate its ability to identify the most stable adsorption configuration on the OC20Dense benchmark. All comparison models are trained on OC20 IS2RE tasks and applied directly without any task-specific retraining, providing a rigorous test of generalization under realistic conditions. For each adsorbate–catalyst system, the model ranks up to 10 candidate configurations, which always include the DFT-relaxed global minimum together with additional structures sampled from the system-specific dense pool. We report three metrics:

- 1) Top- k hit rate: the fraction of systems for which the true global minimum is ranked among the model’s top- k most stable predictions ($k = 1, 2, 3$).
- 2) Success rate (SR): following the OC20Dense protocol, the fraction of systems where the selected configuration lies within 0.10 eV of the DFT-relaxed global minimum energy.
- 3) Inference throughput (Forward samples/s): the number of configurations processed per second, quantifying computational efficiency.

Quantitative screening performance is presented in Table V. Our *Taco* framework dominates all ranking evaluation indicators. It records 14.26% top-one hit rate, 27.34% top-two hit rate, and 38.12% top-three hit rate, surpassing geometry-based EquiformerV2 and text-only CatBERTa. The reference model EquiformerV2 yields corresponding hit rates of 13.40%, 26.20% and 37.03%, while CatBERTa reaches 13.23%, 24.74% and 36.47% respectively. *Taco* also achieves the optimal success rate at 48.83% against all comparison methods. Practical computational throughput further validates application potential. *Taco* processes 309.20 samples per second, running far more efficiently than EquiformerV2 and DimeNet++, which handle 202.54 and 149.13 samples each second. Though CGCNN and CatBERTa achieve higher processing speed, their prediction accuracy falls behind our method.

To demonstrate prediction stability, the statistical distribution of the top-1 hit rate is visualized in Figure 3(c). Our method obtains the highest median value at 13.84%, which outpaces DimeNet++, EquiformerV2, CatBERTa, CGCNN, and SchNet with median scores of 13.69%, 13.57%, 13.23%, 12.43% and 11.78% in sequence. Figure 3(d) further compares model stability measured by interquartile range. *Taco* delivers an

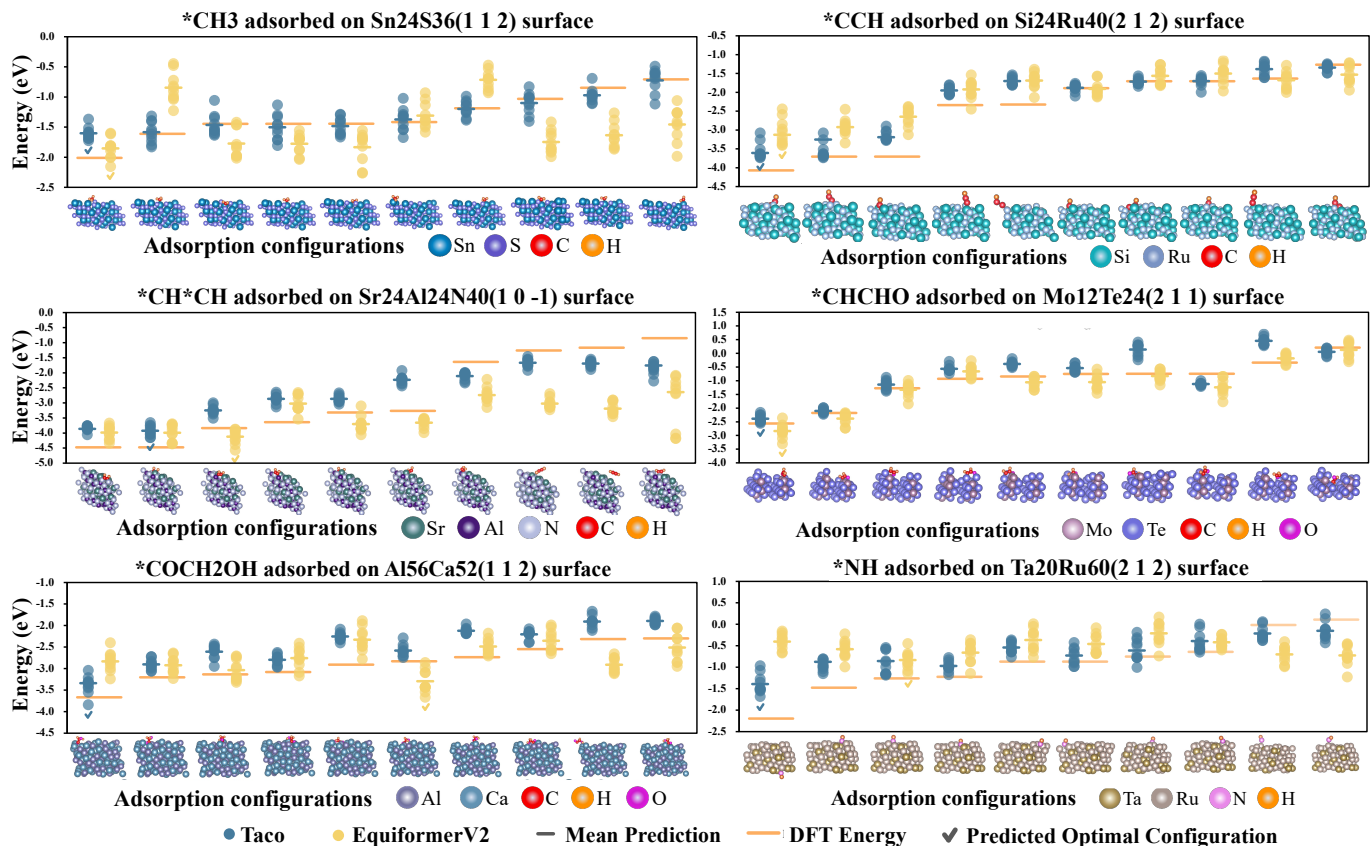


Fig. 5: Predicted adsorption energies for six representative adsorbate-substrate systems. Each panel corresponds to a distinct adsorbate (*CH₃, *CCH, *CH*CH, *CHCHO, *COCH₂OH, *NH) on a different alloy surface. For each system, we compare the energy predictions of *Taco* (blue) and EquiformerV2 (yellow) across multiple candidate configurations, with the DFT-calculated ground-truth energies shown as orange lines.

interquartile range of 1.07, showing reliable performance fluctuation close to stable geometric backbones SchNet and CGCNN with values of 0.97 and 1.02. This stability obviously outperforms CatBERTa and EquiformerV2, whose interquartile ranges stand at 1.38 and 1.54.

Ranking visualization. To further validate the high-throughput screening capability of our model, we evaluated its performance on three distinct adsorbate-catalyst systems with large sets of initial configurations. Figure 4 presents the results for: (1) *CH adsorbed on the K₁₂Mo₃₆S₃₆(2 1 0) surface; (2) *CH adsorbed on the Pt₃₂Ti₁₆Zn₁₆(2 1 1) surface; and (3) *NH₂ adsorbed on the Pt₁₈Ta₁₂Se₄₈(0 1 0) surface. For each system, we include over 40 distinct initial configurations generated via both heuristic and random placement, with their relaxed energies computed via DFT. The right-hand panels illustrate the energy landscape of each system, revealing diverse distributions: a steep energy transition, a gradual slope, and a bimodal-like pattern, respectively. These varied landscapes provide a rigorous testbed for assessing model robustness across different structural and energetic regimes.

The heatmaps compare the predicted rankings from competing methods (CatBERTa, CGCNN, DimeNet++, EquiformerV2, and our *Taco*) against the DFT ground truth, where darker colors correspond to lower predicted/ground truth energies and higher predicted/ground truth rankings. Across all three systems and

across both low- and high-energy regimes, *Taco* consistently reproduces the ground-truth color pattern most faithfully. In particular, our model accurately identifies the dark-colored low-energy configurations, whereas baseline methods often fail to capture these key low-ranking structures, frequently miscoloring them as high-energy (light-colored) states. In contrast, *Taco* not only matches the absolute color intensity but also preserves the relative ordering of configurations, correctly assigning the darkest hues to the most stable structures. These results demonstrate that *Taco* reliably identifies promising candidates across diverse energy landscapes for adsorption configuration discovery.

Screening stability. To evaluate the prediction stability of *Taco*, we compared it with the state-of-the-art EquiformerV2 across six chemically diverse adsorbate-catalyst pairs (Figure 5). Each system features a distinct adsorbate (*CH₃, *CCH, *CH*CH, *CHCHO, *COCH₂OH, *NH) on a different alloy surface, with 10 distinct initial adsorption configurations per system. Both models were independently trained ten times using different random seeds, yielding ten sets of relaxed energy predictions for each configuration.

As visualized, *Taco* exhibits substantially narrower prediction distributions across all six systems, in sharp contrast to the larger variances observed in EquiformerV2. This demonstrates the significantly improved stability and prediction consistency.

Meanwhile, the mean prediction values of *Taco* align closely with DFT-derived ground-truth energies, while EquiformerV2 shows noticeable deviations in some systems. Notably, *Taco* reliably identifies the lowest-energy optimal configuration in all tested systems, whereas EquiformerV2 fails to maintain stable and accurate ranking for critical low-energy structures. These results confirm that *Taco* achieves both high prediction accuracy and stability.

V. CONCLUSION

In this paper, we presented *Taco*, a multimodal framework for efficient and accurate adsorption energy prediction without requiring explicit geometric structures during inference. Motivated by the fundamental trade-off between the predictive accuracy of geometry-based models and the computational efficiency of text-based approaches, *Taco* learns a direct correspondence between textual descriptors and adsorption geometries through a hierarchical cross-modal alignment framework. By combining group-level cluster correspondence discovery with instance-level transport refinement, GCOT establishes fine-grained associations between textual and geometric representations, bridging the modality gap while preserving geometry-related information in the textual embedding space. Extensive experiments on the Open Catalyst benchmark demonstrate that *Taco* consistently achieves competitive or superior prediction performance compared with state-of-the-art geometry-based methods, while significantly reducing inference cost and eliminating the dependence on explicit three-dimensional structures at deployment.

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